

Chenyang LIU

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EDUCATION

Duke University

Durham, USA

PhD in Computer Science, advised by Prof. Matthew Lentz and Prof. Kartik Nayak

Aug. 2024 – Now

Shanghai Jiao Tong University

Shanghai, China

Bachelor of Engineering in Computer Science, GPA: 90.33/100, Rank: 3/86

Sep. 2020 – June 2024

RESEARCH INTEREST

My research interests lie in the areas of distributed systems, blockchain, and security. In particular, I am interested in designing and implementing secure and efficient BFT protocols. Currently, I am working towards design of signature-free BFT protocols for post-quantum era. I am also interested in the design of secure and efficient confidential computing platforms, and the mitigation of side-channel attacks on TEEs.

RESEARCH EXPERIENCES

Research Assistant

Aug. 2024 – Present

Duke University

Durham, USA

Decentralized Proposer-as-a-Service for Ethereum PBS

- **A new side-car protocol for Ethereum PBS:** Formalized the security properties of Ethereum PBS and outsourced the combined functionality of relay and proposer to TEEs.
- **A novel primitive Crab-BB:** a generalized Byzantine Broadcast protocol achieving both censorship-resistance and backed-availability for any application semantic including Ethereum PBS.
- **Implementation and Evaluation:** validate the functionality in a testnet and evaluate the prototype, showing 3% more MEV-earning in DPaaS than relays.

Consensus protocol design for post-quantum era

- **Quintus:** Designed information-theoretic BFT protocols for $n = 5f + 1$ that commit in 2 rounds, matching signature-based protocols with $O(n^2)$ messages and optimistic responsiveness.

Undergraduate Research Assistant

Feb. 2023 – July. 2024

Shanghai Jiao Tong University, advised by Prof. Guoxing Chen

Shanghai, China

Closing Cache Side Channels via Minimal Cache Fingerprints

- **Memory Footprint Minimization:** Designed a software-simulated and fully-associated cache architecture to hide access pattern, enhancing resistance to access-pattern based leakages
- **Compiler Modification:** Instrumented enclave code into 64-bytes blocks by modifying LLVM backend
- **Verification of the OS:** Proposed a novel contract to verify the OS under SGX's threat model

Formally verified privileged enclave as a lightweight and efficient confidential computing platform

- **Case Study and Evaluation:** Implemented three case studies and evaluated the performance of the proposed solution in Penglai platform.

PUBLICATIONS

- Quintus: Two-round Good-case Information Theoretic BFT for $n = 5f + 1$** In Submission
Chenyang Liu, Dahlia Malkhi, Kartik Nayak, Nibesh Shrestha
- DPaaS: Improving Decentralization and Trustworthiness in Ethereum PBS** In Submission
Chenyang Liu, Arnav Jindal, Ittai Abraham, Matthew Lentz, Kartik Nayak
- MINCE: Closing Cache Side Channels via Minimal Cache Fingerprints** Undergraduate Thesis
Chenyang Liu, Guoxing Chen, Jiaping Gui, Haojin Zhu
- A Formal Approach to Multi-Layered Privileges for Enclaves** NDSS 2025
Ganxiang Yang, Chenyang Liu, Zhen Huang, Guoxing Chen, Hongfei Fu, Yuanyuan Zhang, Haojin Zhu

TEACHING/ACADEMIC SERVICES

- Teaching Assistant:** Intro. to Operating Systems (Spring 2025); Operating Systems (Fall 2025)
- Artifact Evaluation Reviewer:** CCS 2027

SELECTED HONORS & AWARDS

- Outstanding Graduate:** Shanghai Jiao Tong University 2024
- Zhiyuan Honor Awards:** Awards to students in Zhiyuan Honor Program (Top 5%) 2021 – 2023
- First Class Network Security Scholarship:** Awards to students who contribute to the research of security 2023
- Academic Progress Awards:** Awards to students who have made great academic progress in one year (3/88) 2022